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Rolls-Royce	ENGINEERING SPECIFICATION	RRES 90009
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Requirements for Design and Development Activities

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1 SCOPE:

1.1 These design process requirements apply to Rolls-Royce Design and Development suppliers authorised by Rolls-Royce to create product definitions using their own design rules and standards. These definitions may range from simple mechanical components to software and complex electronic or mechanical assemblies. The requirements also apply to other product development engineering activities such as design analysis and testing.

1.2 This specification is applicable to the management of design and development suppliers by all businesses within Rolls-Royce and to any inter-company trading within Rolls-Royce group.

1.3 These design process requirements apply to the Organisation and Resource Management, Programme Management, Design and Design Verification of all new and derivative products produced by Rolls-Royce's Design and Development supply chain, including rigs, Special to Product Test Equipment, test facilities and test execution, etc, where required by the purchase order. Requirements for Design and definition of Special to Product Tooling (including Ground Support Equipment (GSE)) are defined in RRES 90059.

1.4 The applicability of each RRES 90009 requirement to the Supplier's products/programmes will be defined by Rolls-Royce and shall be agreed with the Supplier through the development of the Compliance Matrix included within the Design and Development Quality Plan.

2 REASON FOR USE:

2.1 This document places the applicable set of requirements for Design and Development activities (RRES 90009) upon Rolls-Royce's Design and Development supply chain to assure that the requirements of Rolls-Royce's customers and regulators are met.

3 APPLICABLE DOCUMENTS:

The following publications form a part of this document to the extent specified herein. The applicable issue shall be the current issue, unless a specific document issue is specified. When the referenced document has been cancelled and no superseding document has been specified, the last published issue of that document shall apply. Additionally, contact your sector's Rolls-Royce Technical Authority to obtain guidance/approval on action to remove or replace the cancelled reference by a drawing and/or specification change. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence.

3.1 Rolls-Royce Documents:

Available from: https://suppliers.rolls-royce.com or A – Z Index or Teamcenter.

MLC127	List of Nadcap commodity groups and associated Rolls-Royce specifications
RRES 90000	Engineering Control of Manufacturing Source and Method
RRES 90014	Rolls-Royce Aerothermal and Mechanical Rig Test Specification
RRES 90052	Whole Engine Development Test Supplier Requirements
RRES 90057	Materials and Processes to Be Avoided

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RRES 90059 Tool and Fixture Design For Suppliers

Available from: raynesway.standardsoffice@rolls-royce.com

RRES 92000 Control of Manufacturing Source and Method

3.2 Supplier Documents:

Design & Development Quality Plan

3.3 SAE Publications:

Available from SAE, 400 Commonwealth Dr., Warrendale, PA 15096-0001, or **www.sae.org**

AS9102:2014 Aerospace First Article Inspection Requirement

AS9116 Aerospace Series - Notice Of Change (NOC) Requirements

3.4 ANSI Publications:

Available from the American National Standards Institute, www.ansi.org.

TECHAMERICA/ANSI:

GEIA-STD-0005-2 Standard for Mitigating the Effects of Tin Whiskers in Aerospace and High Performance Electronic Systems

3.5 European Standards/ Regulations:

Available online at http://eur-lex.europa.eu/en/index.htm (http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=CELEX:32006R1907:EN:NOT)

EASA Regulations available online at http://easa.europa.eu/regulations

Regulation (EC) No 1907/2006 Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH)

3.6 U.S. Government Publications:

Available from DODSSP, Subscription Services Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111, U.S.A., or online at http://dodssp.daps.dla.mil/

MIL-HDBK-61A Military Handbook Configuration Management Guide

MIL-STD-31000 Technical Data Packages

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4 DEFINITION OF TERMS:

Approval:
Written confirmation that a documented procedure, material or organization, meets the requirements of the appropriate requirement or specification. In all cases where Rolls-Royce has given approval, the supplier remains fully responsible for the quality of the product delivered/affected by the implementation of such actions.

Authorisation:
Written confirmation that a person or organization has been empowered by letter, form or Quality Plan to grant approval. In all cases where authorisation has been given by Rolls-Royce, the supplier remains fully responsible for the quality of the product delivered /affected by the implementation of such actions.

Baseline Configuration:
The approved Design Definition for a product including all functional and physical characteristics necessary for production and production acceptance from which design change control takes place. Configuration of a product, at a specific point in time, which serves as a basis for defining change, for conducting verifications, and for other management activities.

CAGE Code:
Commercial and Government Entity (CAGE) Code. A five-position alphanumeric code (eg, K5086), assigned to organizations that manufacture and/or control the design of parts supplied to a Government Military or Civil Agency or assigned to United States and foreign organizations, primarily for identifying contractors in mechanical interchange of data.

Category A:
The term used to define the retention period of records which relates to products and is ten years after the product type is withdrawn from use (For civil aerospace products this means ten years from withdrawal of the Type certificate and for Military aerospace products this means ten years from the date that Rolls-Royce notifies customers that Rolls-Royce will no longer support the product). Other Category A records need to be retained permanently.

Certificate of Approval:
Rolls-Royce certificate that records the scope of work that a supplier is approved to carry out.

Classified Part:
Parts which have been identified as having failure modes that impact on product safety and/or business performance and hence which required defined controls to be applied to control and assure their integrity through-life. For Aerospace this can be classed as. Critical, Reliability Significant and Business Sensitive parts.

Component:
The level of a system architecture at which system elements are self-contained (i.e. their function arises entirely within the component boundary rather than by interaction between components) and can be implemented to fulfil specified requirements by purchase or produced by engineers who specialise in that item.

Component Maintenance Manual (CMM)
Shows you how to remove, refit, overhaul, inspect, repair and test each item within a component.

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Concession:
The authorisation to use, release or supply a limited quantity of a completed or partially completed product, which does not comply with the specified technical requirements.

Conformance Control Features (CCF):
A measurable feature of a product whose variation has a significant influence on performance or service life. Additional controls are specified to minimise variation in the feature.

Contract:
A written (i.e. not verbal) proposal, tender, bid, offer, offer letter (all where these are capable of acceptance), and any commercial and legally binding agreements, contract or MoU (Memorandum of Understanding) (and their subsequent amendments) with another Party.

Cpk:
The specification range divided by six times the standard deviation of the process Estimates what the process is capable of producing, considering that the process mean may not be centered between the specification limits..

Definition:
In this context, the evolving description of all deliverables necessary to support production and operational support and disposal, of a product package and its individual components throughout its lifecycle.

Design Reviews:
This is a broader term which is the embodiment of the principles of Integrated Product Production Readiness (IPPR) for having a structure to ensure that projects with the potential for significant impacts on supply chains meet customer and business requirements. But it also includes the integration of Technical Reviews within the IPPR process to manage the release of the product definition at various stages or gates.

Design and Development Quality Plan:
A document which details how the Supplier's QMS complies with requirements as described in this document.

Design Technical Data Package:
The required Data items per Appendix A, generated by a Supplier, under the Rolls-Royce Authority.

Deviation Permit.
A Deviation Permit is submitted if, prior to product manufacture, it is known that there is a high probability of unavoidably producing a non-conforming product.

Drawing:
A Product Definition against which an item is procured and the item's configuration is controlled.

Ground Support Equipment:
Storage and Transportation stands, containers, and associated accessories including turnover equipment, blanks, caps etc.

Item:
A non-specific term used to denote materials, parts, equipment, accessories, computer software and documentation.

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Key Characteristics (KC):
An Attribute or feature whose variation has a significant influence on product fit, performance, service life or producibility that requires specific action for the purpose of controlling variation. This definition is further explained in AS9103. This term includes CCF/KCF.

Key Characteristic Features (KCF):
KCFs are parameters of products that are geometric features. KCFs are linked to Product Level Attributes due to their strong influence on Key Characteristics, and will beneficially influence the Product Level Attribute if variation from their nominal value is reduced. KCFs are measureable and can be directly influenced by process control methods at their point of manufacture or assembly.

Major non-conformance:
The non-conforming product adversely impacts one or more of the following: Reliability, Life, Performance or operational behaviour, Maintainability, Interchangeability or Compatibility with mating parts, Material structure or properties, Cosmetic appearance which may be of concern to the Customer, Health and safety, or any other factor defined by the customer as constituting Major.

Minor non-conformance:
The non-conforming product does not adversely impact any of the following: Reliability, Life, Performance or operational behaviour, Maintainability, Interchangeability or Compatibility with mating parts, Material structure or properties, Cosmetic appearance which may be of concern to the Customer, Health and safety, or any other factor defined by the customer as constituting Major.

Non-Conformance (product):
Non-fulfilment of a product's physical attribute or characteristic as described in a specification, drawing, fixed process, or other requirement (contractual, legal, regulatory, technical, customer expectation).

Part:
The physical component or assembly, that is designed, developed and supplied to Rolls-Royce. Typically, a part is identified by a single identification number (part number) in the Rolls-Royce bill of materials.

Product:
The output, created by a Supplier, from the Design and Development process. Products can be hardware, software, documentation, advice, and processed materials or a combination of these categories.

Proprietary Part:
The product of companies other than Rolls-Royce group which are not covered by any Rolls-Royce specification or controlling document, are not standard parts, and are only available from that source.

Repair:
Action taken on either a nonconforming or used product to make it acceptable for the intended use. Repair includes remedial action taken on a previously conforming product to restore it for use, for example as part of maintenance. Unlike rework, repair can affect or change parts of the nonconforming product.

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Rework:
Use of a sequence of approved operations in an attempt to return non-conforming product to a state that conforms completely to the drawing, specification or customer requirement. This may entail the use of repeated standard router operations.

Salvage:
Action on a non-conforming product, which has not been released to the customer, to make it acceptable for its intended use. In some businesses this is known as Production Repair. The Salvage will reduce but not completely eliminate a non-conformance. The purpose of the salvage is to reduce the effect of the non-conformance. Except for Standard Production Repairs, proposed Production Repairs are authorized for use on a one-time basis only by the Design Non-conformance Authority

Self-Approve:
The word 'approved' refers to the formal approval of a specific task / application by Rolls-Royce. Also see Approval and Authorization for further explanation. Where a supplier has been given certain tasks to self-approve this, with the names of the agreed individuals who are responsible for this task shall be recorded in an agreed and configurable document.

Service Bulletin:
A formal document raised to advise customers of the consequence of a modification and their required action. The Service bulletin is approved by the Airframer's regulatory authority. Note this document may not be required for all Rolls-Royce products and businesses

Standard Part:
A part manufactured that is in complete compliance with an established industry or national authority specification (e.g. AeroSpace Defence and Security Industries (ADS), Society of Automotive Engineers (SAE), National Aerospace Standards Committee (NAS) part numbers, Deutsches Institut für Normung (DIN)).

Sub-tier Supplier:
Suppliers not working under a direct purchase order from Rolls-Royce but performing work on Rolls-Royce products at a lower level in the supply chain (via purchase order cascade).

Supplier:
An approved company delivering products to Rolls-Royce against a Rolls-Royce purchase order.

System:
An assembly of elements connected together in an organised way to form a whole, this showing properties of the whole, rather than properties of the elements.

Technical Approval Authority (See Section 6.2.4):
In the context of this document the phrase "Technical Approval Authority" is used to describe the responsibility held by Rolls-Royce in accordance with various regulatory systems or as awarded by customers;

- For Civil Aerospace product, this means the Type Certificate holder or applicant. This is separated into 2 categories:
  - For Engine certified Systems or Components (e.g. JAR- E, CS-E, 14 CFR part 33 etc.) authorisation Technical Authority given by the appropriate Chief Engineer or their delegated authority in Rolls-Royce.

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- For Airframer certified Systems or Components (e.g. JAR-25, CS-25, 14 CFR part 25 etc.) authorisation Technical Approval Authority given by the Air framer or their delegated authority. This may require further negotiation between Rolls-Royce and the Airframer for design approval to the supplier.
- For Energy product this means the Technical Approval Authority
- For Civil Nuclear product this means the Technical Approval Authority
- For Nuclear Submarines product this means Technical Approval Authority
- For Marine, Rolls-Royce is the Design and Technical ApprovalAuthority for its own product
- For Military product this may mean either Design Organisation or the organisation who issues the Certificate of Design.

Technical Checking:
Includes both self-checking and checking of tasks performed by others. It is applicable to all technical tasks and the level of checking may be Light, Detailed or In-Depth depending on a combination of judgments of the probability of errors occurring.

Technical Review:
These are Design, Validation and Verification reviews only. The elements principally concerned with 'ensuring the product delivers the physical and functional customer requirements.

Tin Whiskers:
A phenomenon affecting soldered components, which results in fine whiskers of tin extruding from pure tin surfaces. Tin whiskers are electrically conductive and pose a reliability risk to electronic parts.

Validate:
Confirm through the provision of objective evidence, that the requirements (or specified intended use) have been fulfilled in the intended usage environment.

Verification:
Confirmation, through the provision of objective evidence, that specified requirements have been fulfilled.

Verification Plan:
Scheduled activities and listed data item output to prove and document compliance with each Design Requirement.

Verification Test Procedure:
A step by step methodology defining a specific test procedure, including at minimum the product to be tested, test equipment and calibration, test limits and duration, and criteria for acceptance and failure.

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5 ACRONYMS:

14 CFR	Code of Federal Regulations, Title 14 Aeronautics and Space
ATP	Acceptance Test Procedure
CCF	Conformance Control Feature
DAR	Definition Alteration Request
DCP	Design Change Proposal
DFMEA	Design Failure Mode and Effects Analysis
DR	Definition Review
EASA	European Aviation Safety Agency
FAIR	First Article Inspection Report
FMEA	Failure Mode and Effects Analysis
FMECA	Failure Mode and Effects Criticality Analysis
FRACAS	Failure Reporting Analysis and Corrective Action System
FTA	Fault Tree Analysis
GEIA	Government Electronics and Information Association
IP	Intellectual Property
IPPR	Integrated Product and Production Readiness
IPT	Integrated project Team
KC	Key Characteristics
KCF	Key Characteristics Features
MCRL	Manufacturing Capability Readiness Level
Nadcap	National Aerospace and Defense Contractors Accreditation Program
NOC	Notification Of Change
NPI	New Product Introduction
PFMEA	Process Failure Mode and Effects Analysis
RFA	Request For Alteration
REACH	Registration, Evaluation, Authorisation and Restriction of Chemical substances
SAE	Society of Automotive Engineers
TRL	Technology Readiness Level

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6 REQUIREMENTS:

6.1 Organisation and Resource Management:

6.1.1 The Supplier shall maintain mechanisms, systems or procedures to manage approved signatories for all applicable documentation / products. These mechanisms shall:

- Maintain an active signatory list that is widely available
- Maintain records of past signatories
- Define experience and training requirements for approval signatories

6.2 Programme Management:

6.2.1 Planning, Control, and Resource Management:

6.2.1.1 The Supplier shall engage with Rolls-Royce where required in the development and maintenance of an up-to-date and approved Design and Development Quality Plan (DDQP).

6.2.1.2 The Design and Development Quality Plan shall be reviewed and updated, if required, on any issues where the Certificate of Approval modifies or extends the scope of the Supplier's approval.

6.2.1.3 The Supplier shall demonstrate a structured plan (e.g. project plan with phases, Milestones and agreed dates, sometimes referred to as IPPR landscape within Rolls-Royce) to identify and control actions which are necessary to ensure that projects meet customer and business requirements.

6.2.1.4 The plan shall be appropriately governed with a defined functional representation (Stakeholders) based on the activities defined to meet the requirements.

6.2.1.5 The Supplier shall have a defined escalation route for the resolution of any actions that have been predicted to exceed their timescales.

6.2.1.6 Any open actions recorded through this process shall be reported to Rolls-Royce if they cannot be resolved in the agreed timescales.

6.2.1.7 The Supplier shall demonstrate staged Design and Development process alignment to the Rolls-Royce Technical Review Process. (See Section 6.3.3, 'Technical Reviews', for more requirements reflecting Rolls-Royce's Design Review process).

6.2.1.8 The Supplier shall determine and document all mechanisms, work products, deliverables, etc, that will demonstrate compliance to all Rolls-Royce Technical Requirements.

6.2.1.9 The Supplier shall maintain mechanisms, establish structured project teams, and/or demonstrate practices that consider the cross-functional nature of product design and manufacture throughout the Design and Development lifecycle.

6.2.2 Communication:

6.2.2.1 The Supplier shall maintain appropriate mechanisms for all formal communications between itself and Rolls-Royce. All formal communications shall:

- Be dated
- Be uniquely identified
- Carry applicable part numbers where agreed
- Identify the correspondence initiator

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- Be sent/received through identified points of contact
- Establish and meet agreed targets for communication responsiveness.

6.2.3 Risk Management:

6.2.3.1 The Supplier shall maintain appropriate mechanisms for risk assessment and mitigation to the satisfaction of Rolls-Royce throughout the Product's Design and Development and in-service lifecycles. The Supplier shall collaborate with Rolls-Royce to:

- Define risk assessment criteria and ratings
- Identify, record, prioritise, track, and mitigate project and technical risks
- Assign owners and resolve risk mitigation actions
- Regularly communicate risk assessments and mitigation progress to appropriate Rolls-Royce project teams

6.2.3.2 Risks may be associated with (but not limited to) technology, complexity, obsolescence, materials, environment, application, physical and functional tolerance stack-up associated with the assembly, service and manufacturing requirements, resource availability, information dependencies, supply chain availability, competence, HS&E.

6.2.3.3 The Supplier shall maintain appropriate mechanisms for obsolescence risk assessment and mitigation to the satisfaction of Rolls-Royce throughout the product's Design and Development and in-service lifecycles. The Supplier shall identify, record, prioritise, track, and mitigate all risk of obsolescence.

6.2.3.4 Obsolescence may be associated with (but not limited to) non-availability of standard parts, technological advancement, change in legislation/standards, change in manufacturing processes, compatibility issues, sub-tier suppliers ceasing trade, sub-tier suppliers ceasing production, and sub-tier suppliers ceasing support.

6.2.4 Technical Approval Authority:

6.2.4.1 "Technical Approval Authority" may have a more specific definition according to the industry sector and region applicable to the product or contract. See Definitions.

6.2.4.2 The approval of Rolls-Royce to hold Type Certificates or Technical Approval Authority is granted by a regulatory body or customer. The approval is subject to a programme of surveillance audits by the regulator, customer and / or government agencies, and as such the Supplier shall, at reasonable notice, participate in these audits.

6.2.4.3 Under its approval, Rolls-Royce ensures that all activities carried out by the Supplier are compliant with the appropriate regulations / requirements. Rolls-Royce, at reasonable notice, will regularly assess the competence and systems of the Supplier by oversight audit. The Supplier shall support and facilitate all Rolls-Royce Technical Oversight and any accompanying Regulator Authority Audits.

6.2.4.4 As a consequence of the regulatory systems under which Rolls-Royce operates, it is necessary to identify the responsibilities being undertaken by the Supplier. Issues such as the approval of design changes, the approval of non-conformance, the approval of repairs or salvages, the creation of written instructions (e.g. manuals) the classification of Components, the retention of design data, and the reporting of unsafe conditions are all part of the responsibility held by Rolls-Royce. Rolls-Royce may authorise the Supplier to appoint individuals to conduct activities on its behalf. In this instance, the authorisation must be underwritten by a clear understanding by both parties of the limits of authorisation of the Supplier and by the continued evaluation of the competency of the Supplier to conduct these activities. These activities cannot be further sub-contracted out

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by the Supplier without Rolls-Royce approval. Approval of the Design Technical Data Package cannot be authorised by the Supplier (See Section 6.3.4).

6.2.4.5 Supplier liability for delivery of a Product that meets Rolls-Royce Requirements is in no way diminished by the performance of Technical Oversight/Audit assessments.

6.3 Design:

6.3.1 Requirements Management:

6.3.1.1 Rolls-Royce will develop, formally issue, and update its technical requirements in accordance with planned and agreed arrangements. Their development may occur in stages to allow concurrent design activities. The requirements typically include (but are not limited to):

- Interface (typically, but are not limited to: mechanical, electrical, software)
- Space envelopes/models
- Functional and performance requirements
- Environmental requirements (typically but are not limited to thermal, vibration, shock, cleanliness.)
- Mass, unit cost, reliability, integrity, and life requirements
- Qualification requirements
- Applicable statutory and regulatory requirements
- Applicable Rolls-Royce design policies (e.g. product specific safety policies, safety, Health, Safety & Environment)
- Repair and maintenance costs

6.3.1.2 The Supplier shall provide input and support to the development of the Technical Requirements to enable their satisfactory completion in line with planned and agreed arrangements. The Supplier shall determine:

- Requirements not stated by Rolls-Royce but necessary for specified or intended use, where known
- Applicable statutory and regulatory requirements not stated by Rolls-Royce
- Any additional requirements determined by the Supplier
- A systems evaluation to understand any interfaces, interactions or changes in behaviour of the products and components.

6.3.1.3 The Supplier shall formally review all Rolls-Royce Technical Requirements through a demonstrable process, performed in accordance with planned and agreed arrangements. The Supplier shall:

- Ensure that the requirements are complete (or, where this isn't the case, that there is clear indication and rationale)
- Ensure that the requirements are unambiguous, adequate, and not in conflict with each other
- Confirm that it has the capability to meet the requirements.
- Confirm that it has the means of verifying compliance with the requirements.

6.3.1.4 The Supplier shall communicate any issues or high-risk items with the Rolls-Royce Technical Requirements to Rolls-Royce at the earliest possible opportunity.

6.3.1.5 Changes in requirements or the addition of new requirements during a design project (e.g. as a result of developing the detail of the concept solution, trading requirements

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between contracts, or from gate reviews) will be incorporated by Rolls-Royce in the Technical Requirements at appropriate design stages.

6.3.1.6 The addition of new requirements outside the contractual scope or changes to the requirements after contract award shall be managed using the agreed Rolls-Royce Process as stated in the project contract.

6.3.1.7 The Supplier shall have mechanisms to ensure its Design and Development outputs maintain demonstrable compliance with all Rolls-Royce requirements. The supplier shall:

- Maintain mechanisms to demonstrate complete requirements traceability.
- Maintain mechanisms for the incorporation of new/updated requirements released in accordance with planned and agreed arrangements.

6.3.1.8 Any variances to requirements and change recommendations and rationale shall be documented in ways that are controlled, traceable and retrievable.

6.3.2 Design Methods and Tools:

6.3.2.1 The Supplier shall make full use of agreed design methodologies and techniques throughout their Design and Development process, at system and component level. These shall include, but are not limited to:

- All agreed Design and Engineering disciplines, such as, but are not limited to, Stress, Thermal analysis, Computational Fluid Dynamics
- Design For Quality / Variation (Robust Design)
- Design For Manufacture, Assembly, Aftermarket (Service) and Cost
- Design For Reliability and Safety
- Design for the prevention of foreign objects and contamination being trapped during manufacture, maintenance or repair in such features as, but not limited to, apertures, orifices, sharp bends and corners.

6.3.2.2 Resource managers shall provide suitably qualified and experienced team members to decide which tools to use, including the procedures that they operate, and the standards they adhere to.

6.3.2.3 The Supplier shall agree and perform System / Component modelling and provide the outputs to Rolls-Royce to facilitate wider system integration. The electronic modelling shall be performed and exchanged in an agreed format with Rolls-Royce systems (use of a conversion process is acceptable, if necessary) and include, but are not limited to; mass, centre of gravity and moments of inertia.

6.3.2.4 The Supplier shall optimise the design in order to mitigate the risk of obsolescence. The Supplier shall:

- Identify potential Component, material, tool or process obsolescence during conceptual design.
- Assess and mitigate risks through modular design, use of standard Components/designs, and use of low risk Component materials.

6.3.2.5 The Supplier shall perform a Design Failure Mode and Effects Analysis (DFMEA) for the Product and provide Rolls-Royce complete documentation of the results, in accordance with planned and agreed arrangements. The DFMEA shall include, but is not limited to:

- System / Component identification
- System / Component function

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- System / Component failure mode
- System / Component failure effect on the Supplier's Product function
- Effects of foreign objects and contamination on the system/ component
- Likely rate of occurrence of the failure mode

6.3.2.6 The Supplier shall perform a Process Failure Mode and Effects Analysis (PFMEA) for the Product's production system and provide Rolls-Royce complete documentation of the results (or allow Rolls-Royce access to document for on-site review), in accordance with planned and agreed arrangements.

The supplier shall:

a) Use a cross-function team to establish a PFMEA that includes (but is not limited to) the following:
   - Process identification
   - Process function
   - Process work elements
   - Potential process failure modes
   - Severity (S) – The seriousness of a failure mode in terms of its effect on the Product function
   - Occurrence (O) The likelihood that a given failure mode will happen
   - Detection/Prevention (D) – The likelihood that the failure mode will be prevented or detected
   - Risk Priority Number (RPN) – Severity (S)x Occurrence (O) x Detection (D) = Risk Priority for impact on the product, process and customer
   - Standard scoring criteria
   - Process controls required
   - Identification of all processes specific to the Key Characteristics highlighted by the DFMEA
   - A check against process substantiation or qualification evidence

b) Develop the PFMEA for the production processes identified in the process flow diagram in advance of producing the product

c) Take appropriate corrective action for high RPN"s to reduce or eliminate the chance of the potential failure occurring.

d) Review / update and recalculate RPN"s for the PFMEA when changes are made to product definition, process operating conditions or when non-conformance has been identified

e) Provide feedback to their customer in the supply chain when appropriate risk mitigation cannot be provided

f) Use the cross-function team to develop control plans for the production processes for each product, which defines the controls to be used in advance of producing the product

g) Ensure that the control plan takes into account (but is not limited to) the following elements:
   - PFMEA outputs (see (a) above)
   - Authorised reduced inspection
   - Authorised sample inspection
   - Variation management

h) Ensure that the control plan contents includes (but is not limited to):
   - Part / process number
   - Process name / operation description
   - Product / process characteristics
   - Control method
   - Reaction plan

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i) Review and update control plans when any change occurs affecting product, production process, measurement, logistics, supply sources or PFMEA
j) Maintain a process to review the effectiveness of these controls
k) Maintain records of PFMEA and control plans as category B commencing from the date that the final product was delivered to Rolls-Royce.

NOTE: A single PFMEA and Control Plan may apply to a group or family of products that are produced by the same process at the same source.

6.3.2.7 The Supplier shall demonstrate an agreed product quality planning process, in which findings from DFMEA and PFMEA are fed back into the design process.

6.3.2.8 The Supplier shall review and update its DFMEA and PFMEA in response to any Engineering change and provide Rolls-Royce with the update.

6.3.2.9 A Failure Modes, Effects and Criticality Analysis (FMECA) will be produced for each product. The contract will specify whether Rolls-Royce or the Supplier will be responsible for FMECA production. Where Rolls-Royce produce the FMECA, they will feedback any findings or conclusions to Suppliers.

6.3.2.10 For those design characteristics identified in the DFMEA, PFMEA, and/or FMECA as having significant failure consequences (i.e. the component would be a Classified Part), and where required by Rolls-Royce, the Supplier shall comply with RRES 90000 (RRES 92000 for Submarines Business).

6.3.2.11 All Risks identified by the DFMEA, PFMEA, and FMECA shall be recorded, tracked, and mitigated through inclusion in the Supplier's Risk Management System (See Section 6.2.3, Risk Management).

6.3.2.12 The Supplier shall produce or update its DFMEA and/or its PFMEA in response to known in-service/quality issues or as otherwise agreed between Rolls-Royce and the Supplier.

6.3.2.13 The Supplier shall maintain demonstrable compliance to a Lessons Learnt process (e.g. FRACAS, lessons learnt logs), through which all knowledge and experience gained from previous development and in-service lifecycles are fed back into new designs, design methods, and tools.

6.3.2.14 The Supplier shall record all issues, considerations and design policies affecting the design and maintain records of complete rationale for all design decisions.

6.3.2.15 The Supplier shall perform technical checking of all products and outputs from their Design and Development process/processes. Technical checking is to be considered distinct from 'Technical Reviews' (see paragraphs 6.2.1.2 and 6.3.3) and shall be performed prior to 'Technical Reviews' being conducted. Technical checking is to be performed by Suitably Qualified, Experienced Engineers with defined credentials.

6.3.2.16 The Supplier shall have a process that covers the requirement to conduct technical checking and the authorisation of Suitably Qualified, Experienced Engineers and authorised Engineers as technical checkers.

6.3.2.17 The supplier shall be able to demonstrate a thorough validation strategy so that the operation of the verified product (e.g reliability) is in accordance with the business and product requirements/assumptions when entered into service.

6.3.3 Technical Reviews:

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6.3.3.1 The Supplier shall agree to the applicability of each Technical Review to the scope of its Product's NPI (New Product Introduction) and modification programmes. The use of standard functional checklists where they exist is recommended, provided they contain the minimum content for each gate.

6.3.3.2 The Supplier shall agree to a Technical Review schedule with Rolls-Royce.

6.3.3.3 The Supplier shall agree with Rolls-Royce the roles, accountabilities, responsibilities, deliverables, ownership, attendees, functional checklists, and agenda for each Technical Review.

6.3.3.4 The Supplier shall provide Rolls-Royce with all Technical Review material generally one week in advance of the review itself, recognizing that the complexity of the topic might require a different interval.

6.3.3.5 The Supplier shall hold/attend Technical Reviews in accordance with planned and agreed arrangements.

6.3.3.6 The Supplier shall demonstrate staged Design and Development process alignment to the standard or a project specific Rolls-Royce Technical Reviews Process as stated in the project contract.

6.3.3.7 As part of the scope of the standard functional checklists used in a Technical Review, the supplier shall also refer to the Technology Readiness Levels (TRL) and Manufacturing Capability Readiness Levels (MCRL); Definitions of these are provided in Appendices B and C, respectively.

6.3.3.8 When requested by Rolls-Royce, the Supplier shall keep a formal record of all minutes, actions and salient points agreed upon at each Technical Review, and ensure effective close-out/mitigation at the earliest possible opportunity.

6.3.3.9 Deleted

6.3.3.10 The Supplier shall perform competent technical reviews of all products and outputs from their Design and Development process taking into account the set of design methods and tools discussed in section 6.3.2.

6.3.3.11 Technical reviews shall be configurable and take into account the following stages:

- Project Launch
- Concept(s) Review
- Preliminary Design Review
- Critical design Review
- Review First Parts
- Review Verification and Validation
- Review Early Service and Production Experience
- Review Lessons Learned

6.3.3.12 The Supplier shall show evidence to support the answers from an agreed Technical Review checklist.

6.3.3.13 The supplier shall have some agreed criteria from which to base a decision on whether to pass or fail a Technical Review.

6.3.3.14 The Supplier shall, upon request, give Rolls-Royce full access to details of the design for the purposes of Technical Review, audit, and failure investigation.

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6.3.3.15 Rolls-Royce will give a clear indication of the result of the Technical Review and ensure the result is discussed and agreed prior to the meeting close.

6.3.4 Design Technical Data:

6.3.4.1 The Supplier shall provide a Design Technical Data Package to Rolls-Royce for approval to assure that:

- The Product satisfies the Technical Requirements.
- Any subsequent change to the design still results in a Product that satisfies all technical and regulatory requirements.

6.3.4.2 The Supplier's Design Technical Data Package shall include all the design data items and documentation listed in Appendix A, (unless otherwise stated in the applicable Rolls-Royce Definition or Statement of Work.

6.3.4.3 The Supplier shall submit the initial Design Technical Data Package upon completion or at the next Technical review in accordance with planned and agreed arrangements. Rolls-Royce approval of this package establishes the Baseline Configuration of the Product, forming part of the Type Certificate (or sector equivalent) Bill of Materials.

6.3.4.4 The Supplier shall submit an updated Design Technical Data Package at subsequent technical reviews, in accordance with planned and agreed arrangements. Rolls-Royce approval for Design definitions/ solutions is required before progressing to the next Technical Review.

6.3.4.5 The Supplier shall submit an updated Design Technical Data Package to Rolls-Royce, when specifically negotiated by Rolls-Royce and also when a design change is made (See Section 6.3.11, Control of Design Changes)

6.3.5 Product Safety:

6.3.5.1 Design for Safety:

6.3.5.1.1 Rolls-Royce will identify within the contract whether the product has an impact on safety. Where it does, Rolls-Royce will specify the safety requirements that the product has to meet. The safety requirements will form a subset of the Technical Requirements described in section 6.3.1.

6.3.5.1.2 The supplier shall implement an appropriate safety programme which shall be integrated with the product design and development programme

6.3.5.1.3 A system safety assessment shall be carried out using recognised techniques (e.g. FMECA, FTA) to identify the causes and effects of failures. The assessment shall be recorded in formal documentation which is reported and agreed with Rolls-Royce. Key assumptions shall be recorded. Should any failure modes be identified which have an impact on safety but which are not covered by the specified safety requirements, then these shall be formally reported to Rolls-Royce.

6.3.5.1.4 The DFMEA and PFMEA assessments (see para 6.3.2.5 and 6.3.2.6) shall be used as inputs to the safety assessment.

6.3.5.1.5 The safety assessment shall be used to classify the product and elements within it based upon the consequence of the failure modes. The supplier shall either utilise their own classification scheme or adopt the Rolls-Royce scheme for safety classification. Regardless of the scheme or method used, Rolls-Royce shall approve the classification of the part. The safety classification shall be used to focus effort on reducing safety risk.

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6.3.5.1.6 The product shall be designed such that all of the safety requirements are satisfied. Verification shall be provided for each Safety Functional Requirement. Should the Supplier encounter difficulties in satisfying any safety requirement, this shall be raised with Rolls-Royce at the earliest opportunity so that the implications can be considered and a solution determined. The Supplier shall apply the As Low As Reasonably Practicable (ALARP) principle in developing the product and in satisfying the safety requirements. The Supplier shall demonstrate to Rolls-Royce how the ALARP principle has been applied as part of the safety assessment.

6.3.5.1.7 Any events at any time that question the safety assessment or verification shall be investigated and addressed.

6.3.5.1.8 The safety controls needed to deliver product safety throughout its-life cycle shall be identified and reported and agreed as part of the safety documentation. Types of safety control include:

- Product quality requirements
- Manufacturing requirements
- Commissioning / pass-off test requirements
- Through-life examination, maintenance, inspection and testing requirements
- Component lifetimes
- Operating limits and rules
- Normal and emergency operating instructions
- Requirements for in-service monitoring and feedback

6.3.5.1.9 The supplier shall comply with the agreed Rolls-Royce safety policies as stated in the project contract.

6.3.5.2 Safety Problems:

6.3.5.2.1 The Supplier shall maintain mechanisms for the identification and resolution of all product safety problems. Where there are implications for product safety, they shall be reported to Rolls-Royce at the earliest possible opportunity.

6.3.5.2.2 The Supplier shall report all findings relating to the Product's safe and reliable operation, identified at any stage in the Product's life (from design through to maintenance/overhaul), to the Rolls-Royce Buyer within 24 hours of their discovery. The communication shall be titled "NOTIFICATION OF POTENTIALLY UNSAFE CONDITION".

6.3.5.2.3 The Supplier's interpretation of any finding relating to safety or reliability will be welcomed by Rolls-Royce; however, as the type certificate holder/Technical Approval Authority for the product, Rolls-Royce will make the final judgment on whether there is an actual risk to safe and reliable operation.

6.3.5.2.4 Where Rolls-Royce considers that a risk to the Product's safe and reliable operation exists; the Supplier shall initiate customer protection measures and work collaboratively with Rolls-Royce on corrective actions necessary to eliminate the risk.

6.3.5.2.5 The Supplier shall maintain records of all problems encountered during the Product's development and in-service lifecycle, together with details of the investigation carried out, corrective actions, and their verification.

6.3.5.2.6 The Supplier shall provide all necessary information and assistance to Rolls-Royce, and/or to any third parties identified by Rolls-Royce, to investigate Product Safety findings related to or affecting the supplier's Product.

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6.3.5.2.7 The Supplier shall implement any actions in relation to its Product's design, manufacture, repair, maintenance, or inspection that are considered necessary, by either the Supplier or Rolls-Royce, to resolve any Product safety concern.

6.3.6 Sub-tier Approval and Control:

6.3.6.1 The Supplier shall demonstrably flow down all applicable Rolls-Royce Technical (including regulatory) and contract-specific process requirements to its sub-tier suppliers.

6.3.6.2 The Supplier shall self-approve sub-tier suppliers to create design definitions, using the sub-tier's own design rules and standards, only where formal Rolls-Royce authorisation has been given.

6.3.6.3 If authorised by Rolls-Royce to do so, when the Supplier authorises a sub-tier Supplier to create design definitions of products, using the sub-tier's own design rules they shall demonstrably show they are aligned to the applicable Rolls-Royce, and contract specific process requirements.

6.3.6.4 The Supplier shall maintain complete and up-to-date records of all sub-tier suppliers (used for the contract) together with their approval status, the Product provided, and the organisation and interfaces between the supplier and the sub-tier.

6.3.6.5 The Supplier shall assess the capability of its supply chain, and its compliance to the flowed-down requirements, through an agreed process (including audit).

6.3.6.6 Any data collected during Supplier supply chain assessment shall be made accessible to Rolls-Royce and any regulatory authorities, upon request.

6.3.6.7 The Supplier shall establish and operate agreed controls to manage identified areas of risk (Refer to paragraphs, 6.3.1.4, 6.3.2.1, 6.3.2.4, 6.3.2.11, 6.3.5, 6.3.9.6, 6.3.18.6 and 6.3.2.5) within its sub-tier suppliers.

6.3.7 Export Control, Intellectual, and Other Property:

6.3.7.1 The Supplier shall maintain demonstrable compliance with all applicable Export Control Regulations. The Supplier shall:

- Notify Rolls-Royce of all items, both tangible and intangible, that are deemed to be Controlled Technology (Military or Dual-Use).
- Notify Rolls-Royce of all items, both tangible and intangible, that are U.S. Technology and controlled by ITAR (International Traffic in Arms Regulations) or EAR (Export Administration Regulations) regulations and supply their USML (United States Munitions List) classification or ECCN (Export Control Classification Number).
- Notify Rolls-Royce of any item that may contain, in part, U.S. Technology.
- Ensure that the correct and current Export Licences are in place to allow delivery of the product to Rolls-Royce and/or to other parties.
- Ensure that Export Licences are renewed in adequate time to prevent interruption of the product delivery or technical assistance.
- Ensure that End Use Control is correctly managed.
- Ensure compliance with the U.S Deemed Export Regulations, by liaising and agreeing with Rolls-Royce regarding the nationalities of the Receiving Party Team members.

6.3.7.2 The Supplier shall not incorporate any export controlled items (tangible or intangible) into designs without informing Rolls-Royce and gaining prior written authorisation from Rolls-Royce.

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6.3.7.3 Deleted

6.3.7.4 If Rolls-Royce has funded in whole or in part the development of the technology used by the supplier in the realisation of the design, then the Supplier shall provide Rolls-Royce as early as possible with full details of the design via the Design Technical Data Package.

6.3.7.5 If Rolls-Royce or the Supplier has funded a realisation of the design, based on technology that is the intellectual property of the Supplier, the Supplier shall provide Rolls-Royce with sufficient information, via the Design Technical Data Package, to allow the component to be integrated into the Rolls-Royce system.

6.3.7.6 Any supplier IP provided before any work has started shall be agreed in the contract at the start of a project, to agree what interface information and the extent of this information is to be provided.

6.3.7.7 Contractual obligations may take precedence over the intellectual property requirements outlined above. Rolls-Royce will negotiate any proposed deviation from the Suppliers contractual obligations.

6.3.8 Materials Definition and Control:

6.3.8.1 The Supplier shall produce and control materials definitions so that materials properties used in design are reliably and consistently delivered into service.

6.3.8.2 Materials Definitions shall include the scope of application, condition of supply, and any subsequent processing.

6.3.8.3 The Supplier shall ensure that personnel with agreed credentials approve the materials selection and processing.

6.3.8.4 The Supplier shall agree with Rolls-Royce the material property data for use in Design.

6.3.8.5 The Supplier shall state how incoming material will be verified upon receipt, in terms of whether it conforms to specification and is free from defects.

6.3.8.6 Where castings/forging or Condition of Supply drawings are used, the supplier shall state who is responsible for defining and approving drawings (including non-destructive testing and acceptance criteria).

6.3.8.7 Where castings and forgings are used, they shall be controlled by drawings within the Supplier's configuration management system.

6.3.8.8 The Supplier shall document how Components are to be validated for mechanical properties, grain flow and size, and defects.

6.3.9 Materials and Manufacturing Processes to be avoided:

6.3.9.1 The Supplier shall comply with any Rolls-Royce standards or policies on restricted chemical substances, material combinations, and processes as defined by RRES 90057.

6.3.9.2 The Supplier shall provide sufficient information/data so as to enable Rolls-Royce to comply with its own obligations under Regulation (EC) No 1907/2006 - REACH.

6.3.9.3 Rolls-Royce may require suppliers to provide information about the constituent substances within the Component being supplied.

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6.3.9.4 The Supplier shall, when they have an obligation under Regulation (EC) No 1907/2006 – REACH, comply with the requirements so as to ensure continuity of supply.

6.3.9.5 Non-disclosure agreements shall not exempt the Supplier or sub-tier Supplier from REACH, RoHS (Restriction of the use of certain Hazardous Substances in electrical and electronic equipment), WEEE (Waste Electrical and Electronic Equipment) or any other legal obligations.

6.3.9.6 Specifically for electronic and electrical systems (to manage Product safety risks associated with use of lead-free solder and lead-free component surface finishes), any use of non-lead solder electronic products supplied to Rolls-Royce for use in end-systems shall be specifically agreed with Rolls-Royce.

Note: The remainder of this section (6.3.9.7 – 6.3.9.9) applies to Aerospace Products only:

6.3.9.7 Suppliers of electronic Components supplied to Rolls-Royce for use in end-systems shall document and agree a Control Plan for lead-free finishes in accordance with AIA (Aerospace Industries Association) standard GEIA-STD-0005-2. This Control Plan shall include:

- Supply chain monitoring controls
- Mitigations used to prevent tin whisker formation or any other known undesirable characteristics
- Evidence for mitigation effectiveness

6.3.9.8 Suppliers of electronic and electrical product shall adhere to level 2B controls defined in GEIA-STD-0005-2. Rolls-Royce reserves the right to selectively increase the control level based on customer requirements or function integrity requirements.

6.3.9.9 Engine control and protection units shall be inspected for the occurrence of tin whiskers, or any other known undesirable characteristics, to a process agreed in the Control Plan.

6.3.10 Control of Special Processes:

6.3.10.1 For Aerospace applications, the Supplier shall consider all those processes listed in Rolls-Royce specification MLC127 (as a minimum) as special processes.

6.3.10.2 The Supplier shall validate and ensure that special processes are controlled through agreed procedures which have Nadcap accreditation and have been approved by identified personnel with defined credentials.

6.3.11 Control of Design Changes:

6.3.11.1 After the baseline configuration is established, any change to any document that defines the configuration of an item is considered a “Design Change” and shall be managed and controlled in accordance with Appendix D. Only those individuals employed by the supplier who have been assessed for competence and endorsed by the applicable Rolls-Royce Technical Approval Authority may self-approve Design Changes. Where no individuals employed by the supplier have been endorsed by Rolls-Royce the supplier shall obtain Rolls-Royce approval for design changes prior to embodiment of the change in delivered product. Such changes are conducted in accordance with Appendix D.

6.3.11.2 A Supplier may seek such self-approval authorisation through the Rolls-Royce point of contact. Authority is granted to named individuals who have been assessed for

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competence by Rolls-Royce and the Supplier. Rolls-Royce shall retain the right to veto the selected people.

6.3.11.3 The Supplier shall assure that Design Changes conducted by sub-tier suppliers are carried out in accordance with the requirements of 6.3.11.

6.3.12 Non-Conformance:

6.3.12.1 All non-conformances that affect Rolls-Royce Engineering Requirements have to be reported to Rolls-Royce for sentencing by Rolls-Royce.

6.3.12.2 Any of the following delegations by Rolls-Royce to the Supplier may not be further delegated by the Supplier to its own sub-tier Suppliers without the explicit approval by Rolls-Royce.

6.3.12.2.1 Minor non-conformance that does not affect the Rolls-Royce Engineering Requirements must be documented and can be sentenced by appropriately qualified people designated by the Supplier, and Rolls-Royce does not need to approve these people by name or program - but the Supplier must be approved by Rolls-Royce, by Vendor Code, to do this. The non-conformance records must be maintained in accordance with record retention requirements, and made available if required for review by Rolls-Royce.

Note - When the clause FAR.54.246-2, Inspection of Supplies – Fixed Price, applies, these “minor non-conformances” must be presented by the suppliers to the delegated US Government agency for acceptance prior to delivery

6.3.12.2.2 Minor non-conformance that affects the Rolls-Royce Engineering Requirements may be delegated by the Rolls-Royce Chief Engineer only if the following conditions are met:

- The Chief Engineer has provided authorization for specific product and applications to individuals by name within the supplier to perform the sentencing. The level of authorization for an individual will be defined within an agreed delegation document.
- Where Rolls-Royce customer approval or traceability is a requirement, the Chief Engineer may only delegate non-conformances that the customer has delegated to Rolls-Royce. i.e., Rolls-Royce can only delegate supplier non-conformance self-approval where customer approval is not required.
- If the product is eligible for other applications outside the initial Chief Engineer’s remit, the Chief Engineer for these other applications must also provide authorization
- The non-conformance is repeat minor. Therefore, the initial non-conformance must always be sentenced by Rolls-Royce, but subsequent non-conformance sentencing can be delegated.
- The parts are not classified as critical by Rolls-Royce.
- The non-conformances are not deviations

6.3.12.3 Any Rolls-Royce delegations are rescinded in cases where multiple non-conformances are present on a given part where the supplier does not have delegation for all non-conformances present. In these cases, the supplier must report all non-conformances to Rolls-Royce Engineering requirements (even those with delegation) to Rolls-Royce for sentencing by Rolls-Royce

6.3.12.4 Major non-conformance and all deviations must always be sentenced by Rolls-Royce.DP8D

6.3.13 Control of Salvage Schemes:

6.3.13.1 The Supplier shall submit salvage schemes for the processes used for reclaiming Non-

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conforming Components or features during manufacture to Rolls-Royce for approval. The salvage scheme shall:

- Define tolerance limits, processes, and procedures for the reclaimable features during salvage.
- Include processes and specifications for inspection and verification of the salvage.

6.3.13.2 Where formal Rolls-Royce authorisation has been given, (application made through Rolls-Royce point of contact, authority granted by the applicable Rolls-Royce Technical Approval Authority see section 6.2.4), named and authorised individuals within the Supplier shall self-approve Salvage Schemes and Salvage Scheme Amendments. The level of authorisation for an individual will be defined within an agreed delegation document.

6.3.14 Design Source Changes:

6.3.14.1 The Supplier, and its sub-tiers, shall manage and control Design Source Changes in accordance with Appendix D using the applicable Rolls-Royce process.

6.3.14.2 Where authorisation has been granted (application made through Rolls-Royce point of contact, granted by the applicable Rolls-Royce Technical Approval Authority see section 6.2.4), named and authorised individuals within the Supplier shall self-approve Design Source changes. The level of authorisation for an individual will be defined within an agreed delegation document.

6.3.15 Engineering Control of Manufacturing Source and Method:

6.3.15.1 The supplier shall have in place a system for identifying which manufacturing processes are critical to the performance, integrity or reliability of the part and are subject to the design and development Supplier's Engineering approval of any change to source or method.

6.3.15.2 Where additional control is required for Rolls-Royce Fixed Process Approval, Rolls-Royce will require RRES 90000 (RRES 92000 for Submarines Business) by call out on the Engineering

6.3.15.3 Source Control Drawing.

- The supplier shall notify Rolls-Royce of any such changes that require Engineering approval, using the Eng. Control of Manufacturing and Repair Source and Method Form 2 at the Rolls-Royce business requirements location. These changes shall be defined in the DDQP when RRES 90000 (RRES 92000 for Submarines Business) is required.
- The supplier shall flow this requirement down to sub-tiers if operations which require Engineering approval of any change to source or method are outsourced.

6.3.16 Component Proving:

6.3.16.1 The supplier shall demonstrate that the manufacturing meets the component/part definition requirements (e.g. using a FAIR process in compliance with AS9102).

6.3.16.2 The Supplier shall identify and record all manufacturing/assembly/process Key Characteristics in applicable process control documentation of its Products, communicate them and their values to Rolls-Royce for approval.

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6.3.16.3 The Supplier shall ensure its Component's Key Characteristics are successfully communicated to its manufacturing and assembly departments.

6.3.16.4 Key Characteristics shall be managed using the agreed Rolls-Royce process as stated in the project contract.

6.3.16.5 All features of the component or assembly shall have their effect on product functionality considered using DFMEA. This will identify Key Characteristics to be monitored.

6.3.16.6 The Supplier shall rectify any instances where the Cpk of Key Characteristics falls below 1.33 through effective root cause analysis and completion of corrective actions.

6.3.17 Non-Destructive Evaluation:

6.3.17.1 The Supplier shall specify the criteria used for determining whether Non-Destructive Evaluation is an acceptance requirement, and agree the requirement with Rolls-Royce.

6.3.17.2 The Supplier shall specify the criteria used for determining the type of Non-Destructive Evaluation to be used, and agree the requirement with Rolls-Royce.

6.3.18 Supplier Use of Finished Parts:

6.3.18.1 Finished Parts include:

- Parts designed by the Supplier but made by its sub-tier.
- Parts designed and made by a sub-tier specifically for a Supplier.
- Proprietary or Standard parts, whether purchased or made by the supplier.

6.3.18.2 The Supplier shall obtain approval from Rolls-Royce for the use of proprietary parts in its design.

6.3.18.3 The Supplier shall ensure that any other applicable quality requirements are applied and demonstrated upon receipt of purchased Finished Parts.

6.3.18.4 The Supplier shall manage configuration control and Design and Development changes of purchased Finished Parts in accordance with Appendix D.

6.3.18.5 The Supplier shall perform additional risk assessments associated with the obsolescence, technological development, manufacturing, legislation, and standard of purchased Finished Parts.

6.3.19 Note: This section which was previously titled Classified parts has now been merged with paragraph 6.3.5, Product Safety. Retention of the clause number 6.3.19 remains as it is referenced in several other documents.

6.3.20 Serialisation of Parts:

6.3.20.1 The Supplier shall create and maintain records to enable tracing of serialised parts to their sub-tier Supplier from any point in the Supplier's supply chain.

6.3.21 Repair Schemes:

6.3.21.1 The Supplier shall submit repair schemes to Rolls-Royce prior to their implementation/incorporation into the applicable Component Maintenance Manual. The repair schemes shall:

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- Define tolerance limits, processes, and procedures for reclaimable features during repair and overhaul (in-service).
- Include processes and specifications for inspection and verification of the Repair.
- Include all necessary instructions and limitations.

6.3.21.2 The repair schemes shall comply with Rolls-Royce's formatting and data requirements.

6.3.21.3 The Supplier shall produce and formally issue repair schemes in response to defects/faults discovered in-service.

6.3.21.4 For any changes to source or method associated with repair schemes for classified parts, RRES 90000 (RRES 92000 for Submarines Business) shall apply.

6.3.21.5 Where formal Rolls-Royce authorisation has been given, (application made through Rolls-Royce point of contact, authority granted by the applicable Rolls-Royce Technical Approval Authority see section 6.2.4), named and authorised individuals within the Supplier self-approve an agreed category of repair scheme amendments and submit them to Rolls-Royce for inclusion in the Maintenance Manual. The agreed level of self authorisation for an individual will be defined within an agreed delegation document.

6.3.22 Document and Data Retention:

6.3.22.1 Any Rolls-Royce process that has been agreed in the project contract shall apply to the Supplier and its sub-tier Suppliers.

6.3.22.2 The Supplier shall treat all Design Technical Data as category A records unless otherwise agreed by Rolls-Royce.

6.3.23 Disposal:

6.3.23.1 The Supplier shall dispose of all Components, or equipment used in the Design and Development process and/or any life-expired Components returned from in-service, appropriately at the end of their lifecycles. The Supplier shall:

- Prevent unauthorised salvage.
- Minimise environmental impact.
- Satisfy national or international Health, Safety, and Environmental regulations.
- Comply with any national security regulations.
- Safeguard Rolls-Royce Intellectual Property.

6.3.23.2 The Supplier shall document appropriate disposal procedures and instructions within the applicable Component Maintenance Manual.

6.3.23.3 The Supplier shall agree the disposal procedures with Rolls-Royce for any items subject to national security classifications and National/International regulations.

6.3.24 Configuration Control:

6.3.24.1 The Supplier shall follow a defined process for Configuration Identification and Management, in accordance with MIL-HDBK-61A, ANSI/EIA-649-A 2004 or similar.

6.3.24.2 The configuration of the design shall be managed and traceable from the Supplier Item Drawing through the Supplier's entire Drawing package including Sub-tier Supplier Drawings.

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6.3.24.3 All Supplier's, including Sub-tier Supplier's, responsible for each of their products shall be identified, using an agreed configuration controlled database, preferably by reference to the agreed CAGE (Commercial and Government Entity) code for that entity, or by reference to another agreed configuration system that contains this information. Supplier information associated with any CAGE code can be obtained from the following web-site:

https://cage.dla.mil/Search

Please note that the web-site address may change from time to time; these changes are outside the control of Rolls-Royce.

6.3.24.4 Any change to form, fit or function, shall be referred to Appendix D to determine if a change of part number is required.

6.3.24.5 All drawings associated with the design or manufacture of any delivered part shall be issued prior to part delivery or use in qualification tests.

6.3.25 Failure Reporting and Corrective Action System (FRACAS):

6.3.25.1 The Supplier shall report all component failures to Rolls-Royce during its development, build or production pass off testing and any in-service incidents they are made aware of.

6.3.25.2 The Supplier's Quality procedures shall support the execution of the FRACAS system at Rolls-Royce and shall contain at least the following documentation to be delivered:

- Initial failure report,
- Root cause report,
- Close-out report containing corrective action measures including an agreed risk analysis.
- Furthermore, a status reporting shall be provided throughout the failure investigation phase.

6.3.25.3 The Supplier shall provide the root cause report and the close-out report to Rolls-Royce.

6.3.26 Service Bulletins:

6.3.26.1 If the Supplier were to raise a Service Bulletin (SB) for a change/inspection to one of its components, Rolls-Royce can either use their SB as an attachment to the Rolls-Royce SB or incorporate its content into the Rolls-Royce SB. Only the Rolls-Royce approved SB can be used as "Approved Design Data".

6.3.26.2 The Supplier's SB format shall be in accordance with ATA Spec 100 requirements and has to be agreed with the relevant Rolls-Royce 'Engineering for Services' department or equivalent.

6.4 Design Verification:

6.4.1 The Supplier shall verify the output of the Design and Development Process has met the Design and Development input requirements (including. Rolls-Royce, regulatory, Supplier generated) through appropriate mechanisms/processes. These may include (but are not limited to):

- Verification by Analysis (i.e. technical evaluation through models, simulations, algorithms, and drawings).

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- Verification by Comparison (i.e. assess that the new definition is similar or identical in design and manufacturing process to another definition that has been previously qualified to equivalent or more stringent requirements).
- Verification by Inspection (i.e. physical examination of hardware, documentation, to verify compliance with requirements).
- Verification by Test (i.e. testing of the actual operation of definition under expected operational environments to evaluate performance).
- Verification by Demonstration (i.e. demonstration that the definition can perform its design functions under specified scenarios).

6.4.2 The Supplier's verification activities shall provide evidence of the mitigation of key technical and product safety risks.

6.4.3 Any intermediate verification activities performed during the Design and Development process shall form part of the draft Design Technical Data Package submitted to Rolls-Royce at the defined Technical Reviews, in line with planned and agreed arrangements.

6.4.4 Where test is used for verification purposes, the test shall be defined and recorded to enable audit and reproducibility. The Supplier's test verification mechanisms / processes shall meet those applicable requirements defined in RRES 90014 and RRES 90052. These requirements include (but are not limited to) the following:

- Calibration
- Test facility configuration control
- Verification planning
- Verification procedures/methods
- Verification inspection and data acquisition
- Verification witnessing and evidence
- Verification documentation

6.4.5 Where analysis is used for verification purposes, the following shall be defined and recorded to enable audit and reproducibility:

- Verification plan including the approach, method, test cases and acceptance criteria to be analysed
- The type and versions of tools, software, and software platforms used
- Tool/software verification plans and results where the tool or software is customised or bespoke for the purpose
- Source data used where relevant (e.g. materials data for stress analysis)
- The definition/configuration of the product analysed
- The results of the analysis
- Pass/fail conclusions where applicable

6.4.6 Certification Testing:

6.4.6.1 The Supplier shall have appropriate mechanisms for performing necessary certification activities to ensure the Product complies with applicable regulatory requirements. The supplier shall:

- Agree with Rolls-Royce which certification activities require Rolls-Royce test facility approval.
- Where applicable, use a Rolls-Royce approved test facility if contracting or outsourcing certification activities.

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6.4.6.2 Rolls-Royce approval of a test facility may take the form of an on-site assessment of the test facility (prior to test) and/or approval as a Rolls-Royce supplier.

6.4.6.3 Where test is used for certification purposes, the same requirements called out in paragraph 6.4.4 shall apply.

6.4.7 Acceptance Test:

6.4.7.1 The Supplier shall prepare an Acceptance Test Procedure (ATP) for each design definition. The ATP shall:

- Serve as the standard for acceptance of new or repaired Components.
- Include pass/fail criteria for new and service Components.
- Be approved by Rolls-Royce prior to being used.
- Demonstrate the functional accuracy of the Component.

6.4.7.2 Every Component produced by the Supplier shall satisfactorily pass the ATP prior to being accepted by Rolls-Royce.

7 NOTES:

7.1 Revision bars indicate changes from the previous issue of this specification and are located in the left margin.

Issue	Change (ESC) Number	Change Summary (in descending order)	Date
H	ESC500026366884	Revised scope to reference RRES 90059 for design requirements for Special To Product Tooling including Ground Support Equipment. Referenced RRES 92000 in addition to RRES 90000 for Submarines application throughout. Link to EASA Regulations added . Added definitions: Ground Support Equipment, Major non-conformance, Minor non-conformance, and non-conformance Revised Section 6.3.12 regarding non-conformance RFS 1158 Duplication between Section 6.3.3.9 and section 6.3.3.15 removed. Revised Appendix D, section D4 to include instructions about which notification of change form to use depending on the Rolls-Royce site that commissioned the design-make part RFS 1174 Revised Section 6.3.12.3 to recognize the	17 May 2016

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Issue	Change (ESC) Number	Change Summary (in descending order)	Date
		impact of FAR.54.246-2 on reporting of non-conformance RFS 1180 6.3.2.6 Process failure mode and effects analysis strengthened, 6.3.15 Updated to require suppliers to notify Rolls-Royce of changes to manufacturing source and method that require Engineering approval RFS 1185 Appendix E and all references to it removed	
G	ESC500022712413	New Definitions for Approval, Authorisation, Category A, Cage Code, Classified Part, Certificate of Approval, Component, Component Maintenance Manual, Cpk, Deviation Permit, Conformance Control Features (CCF's): Key Characteristics (KC's) Key Characteristic Features, Product, Salvage, Self-Approve, Service bulletin, Standard Part, System, Technical Checking, Technical Review, and Validate. Work Product definition removed and replaced with just 'Product' FAR term replaced by 14 CFR Section 1 – tooling removed from scope Following EASA audit and August Ballot on RRES90009 in 2014 number of updates made to para 6.2.1.1, to 6.2.1.7 and 6.2.1.9. KCF term added to Paragraph's 6.3.2.6, 6.3.16, Ballot Comments: Appropriate replaced with agreed affects paragraphs 6.2.2, 6.3.1.5, 6.3.1.6, 6.3.2.1, 6.3.2.2, 6.3.2.3, 6.3.2.6, 6.3.2.7, 6.3.2.10 - 6.3.2.12, 6.3.2.15, 6.3.2.17, 6.3.3.3, 66.3.6.5, 6.3.6.7, 6.3.8.3, 6.3.10.2, 6.3.16.3, 6.3.22.1, 6.3.23.2, 6.3.24.3, 6.4.1, 6.4.6.1 Manufacturing, materials added to 6.3.2.13 Para 6.3.3.1 updated to add the use of functional checklists Para 6.3.3.4 updated to clarify when definition review material provision. Earliest possible opportunity added to paragraph 6.3.3.8, 6.3.14, 6.3.7.4 New paragraph 6.3.2.15 added to define technical checking. Paragraph 6.3.4.2 and 6.3.4.3 updated to clarify when a Design Technical Data Package is required. Para 6.3.6.3 updated to show sub-tier	20 Jan 2015

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Issue	Change (ESC) Number	Change Summary (in descending order)	Date
		demonstrate alignment of RR contract or process requirements. 6.3.7 Export Control added to title Paragraph 6.3.7.6 – New Requirement to agree IP at the start of a project. To ensure suppliers have an export with groups other than just Rolls-Royce - 'And/or other parties added to 6.3.7.1 6.3.7.3 moved to 6.3.3.14 Update to paragraph 6.3.10.2 and 6.3.10.3 removed to firm the requirement on approvals and authorization for special processes. 6.3.12.3 – new requirement for undeclared non-conformances added Para 6.3.22.1 updated to clarify which RR process are agreed. Product safety risk added to Para 6.3.22.1. – 6.4.7.2 developed by the supplier add to para. RFS 1078	
G	ESC500022712413	– Paragraph added (6.2.1.5) to ensure that the Design and Development Quality Plan and a Supplier's Approval certificate were aligned – RFS 1047	20 Jan 2015
G	ESC500022712413	The following paragraphs have been removed or amended as part of a rationalisation exercise of RRES90009: – , 6.1.2, 6.1.3, 6.1.4, 6.1.5, , 6.2.1.4, 6.2.1.5, 6.2.1.6, 6.2.1.7, 6.2.1.8, 6.3.1.5, 6.3.1.10, 6.3.2.8, 6.3.2.10, 6.3.2.15, , 6.3.3.6, requirements for completion of design reviews from para 6.3.3.8, 6.3.4.3, 6.3.4.7, , 6.3.5.8, , 6.3.7.2, 6.3.7.4 and the requirements for NDAs, and IP protection, , 6, , , 6.3.13.3, 6.3.13.4, , , 6.3.16.1, , 6.3.19.1, 6.3.19.2, 6.4.6.3, removal of previous history changes prior to Rev F, Appendix D. – RFS 1042	20 Jan 2015
G	ESC500022712413	New New Paragraphs/Requirements added 6.3.25 Failure Reporting and Corrective Action System (FRACAS) 6.3.25.1 The Supplier shall report all failures during development, build or production pass-off test and in-service of parts to Rolls-Royce . 6.3.25.2 The Supplier's Quality procedures shall support the execution of the FRACAS system at Rolls-Royce and shall contain at	20 Jan 2015

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Issue	Change (ESC) Number	Change Summary (in descending order)	Date
		least the following documentation to be delivered: • Initial failure report, • Root cause report, • Close-out report containing corrective action measures including agred risk analysis. Furthermore, a status reporting shall be provided throughout the failure investigation phase. 6.3.25.3 Th,e Supplier shall provide the root cause report and the close-out report to Rolls-Royce. 6.3.26.3 Service Bulletins 6.3.26.1 If the Supplier were to raise a Service Bulletin (SB) for a change/inspection to one of its components, Rolls-Royce can either use their SB as an attachment to the Rolls-Royce SB or incorporate its content into the Rolls-Royce SB. Only the Rolls-Royce approved SB can be used as "Approved Design Data". 6.3.26.2 The Supplier's SB format shall be in accordance with ATA Spec 100 requirements and has to be agreed with the relevant Rolls-Royce 'Engineering for Services' department or equivalent. – RFS 1041	
G	ESC500022712413	– Change the definition of "Classified Part" to ensure this includes parts recognised as playing a key role in product safety or performance. – Para 6.3.1, 6.3.4.9, 6.3.4.10, 6.3.5, 6.3.5.2, 6.3.9.6 – replace requirement for product integrity with product safety requirements. – Para 6.3.2.9 – change to requirements for FMECA to mandate their use for each product and for the contract to indicate whether RR or the Supplier is responsible for FMECA. – Para 6.3.5 change the title from "Integrity of Design" to "Product Safety". Section now split into 2 parts, 6.3.5.1 – Design for Safety and 6.3.5.2 Safety Problems. – Para 6.3.5.1.1– amend to indicate the	20 Jan 2015

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Issue	Change (ESC) Number	Change Summary (in descending order)	Date
		RR will identify in the contract whether the product has an impact on product safety and that RR will specify any safety requirements, as required. – Para 6.3.5 - Add new requirements (6.3.5.1.1 to 6.3.5.1.8) to indicate the requirements to satisfy all safety requirements and related verification evidence and to ensure any assumptions related to safety are clearly identified. – Removal of Para 6.3.19 Para as now included in 6.3.5. – Para 6.3.20.2 – add a reference to RRES 900005 – Para 6.4.2 – add mitigation of product safety in addition to technical risks. – RFS 1039	
G	ESC500022712413	– Para 3.1 reference RRES90052 added – Para 6.1.5 removed as covered in ISO 9001 4.1 (e), (f), 8.2.2, 8.5.1 – Para 6.2.1.2 added "Design reviews" into the scope of schedules included in the Programme Management Plan. – Para 6.3.2.1 – changes to endorse/clarify Rolls-Royce Design Methods and Tools. – Para 6.3.2.2 – changes to endorse/clarify Rolls-Royce Design Methods and Tools. – Para 6.3.3 and subsections that refer to Technical reviews (6.3.3.6 to 6.3.3.13) updated to reflect latest practices for carrying out Rolls-Royce Design reviews. – Para 6.3.3.6 – updated to reflect the removal of appendix D and give supplier choice of following a standard or project specific Design Review – Para 6.3.3.7) – amended to reference the Technical Readiness level on the Design review checklists rather than the DR gate checklists. – Added new paras 6.3.3.10 to 6.3.3.13 related to acceptance and completion criteria for a design review. Old section 6.3.3.10 was removed as it was related to the result of a Definition and the requirement has been superceded by the new requirements for Design reviews. – Paras 6.3.13.3 and 6.3.13.4 related to the requirement to report to Rolls-	20 Jan 2015

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Issue	Change (ESC) Number	Change Summary (in descending order)	Date
		Royce all non-conformances on any Component classified 'Critical' or 'Sensitive' and all non-conformances on any identified Key Characteristics or agreed manufacturing Conformance Control Features and to investigate the root cause of all non-conformances have been moved to section 6.3.12 as part of the requirements for non-conformance, rather than the requirements for salvage schemes. – Paras 6.3.12.4 – new requirement on the use of coatings with non-conformances. – RFS 1038	
G	ESC500022712413	– 6.3.3.8: When requested by Rolls-Royce, the Supplier shall keep a formal record of all minutes, actions and salient points agreed upon at each Technical Review, and ensure effective close-out/mitigation at the earliest possible opportunity. – RFS 1012	20 Jan 2015
F	ESC500019316657	– Para 1.3 added: test facilities and test execution – Para 3.1 added RRES 90052 – Section 4: Revised Drawing definition and added Hand Dressing and Rework definitions – New para 6.3.8.9: instructions for hand dressing – Para 6.3.19.5 – ref added to RRES 90000 – 6.3.24.3: update BINCS CAGE code link – Para 6.4.4: added RRES 90052 and new list item about test facility configuration control – Para 6.4.6.3: added RRES 90052 and new list item about test facility configuration control	17 Sep 2013
E	ESC500018511272	– Amended to remove reference to RRDS 11001 in Sections 3.1, 6.3.9.1 and 7.1. In 3.1 and 6.3.9.1 replaced RRDS 11001 with RRES 90057.	29 May 2013
D	ESC500017816588	– Expanded description of Design/Technical Authority to include reference to Engine and Airframer certification plus include work product in definition of terms section. – New Acronym, for STEP. – Word change to supplier material selection to align with description in RRDS 11001. – Re-wording on the Control of design change/non-conformance process for	19 Feb 2013

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Issue	Change (ESC) Number	Change Summary (in descending order)	Date
		suppliers. – One additional requirement to reflect how DFMEA's are used to generate CCF's and the monitoring of these. – Expansion on repair schemes for self approval. – Swapped 7.1 and 7.2 and restored 7.2 to marginal indicia per template. – Appendix A – removed data items "Definition Reviews and First Article Inspection Reports" Separated data items DFMEA and PFMEA.	
C	ESC500015451806	Removed "Aerospace" from front page header.	6 Dec 2011
B	ESC500014893127	Removed all references to SABRe documents where appropriate, Added CAGE code references, Appendix A, added data item codes, clarified data requirements and defined the Supplier Item Drawing. Changes to Appendix D to bring in line with GQP C.4. General updates to improve readability, understanding and cross sector language.	16 Aug 2011
A	ESC500012436415	General update following first release, but including gaps identified through comparison with regulatory requirements, addition of section on configuration control, update of Appendix E following issue of AS9016, and restructuring of sections 6.4 and 6.5.	4 Feb 2010
Initial	---	Initial release of document to formalize Rolls-Rolls requirements for Design & Development activities and supersede EDS 733 EDNS04000002606	25 Mar 2008

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APPENDIX A – DESIGN TECHNICAL DATA PACKAGE

Data Topic	Data Item / Documentation	Notes
Parts List	A – Parts List	Per ASME Y14.34 Assembly, Hardware, and/or Software. A parts list that is integral with the drawing is also acceptable.
Design Definition	B – Installation Drawing(s), C – Assembly Drawings(s), K – Supplier Item Drawing (See notes) 3D – 3D Computer Aided Design (CAD) Model	Unless otherwise specified by Rolls-Royce, drawing(s) deliverable to Rolls-Royce, are to be per MIL-STD-31000 Type 2D, Production Level, in electronic and hardcopy format. Assembly Drawing(s) are required for each Configuration Item (CI) i.e. where an Operator, or Rolls-Royce, has the capability to disassemble the unit into more than one replaceable part or assembly (LRU - Line Replaceable Unit). The Supplier Item Drawing is the top level production drawing of an item that is being supplied to Rolls-Royce. 3D CAD Models of the Component are to be per MIL-STD-31000 Type 3D, STEP (STandard for the Exchange of Product (model data)), unless otherwise specified by Rolls-Royce.
Material Definitions	D – Material List	See Section 6.3.8
Sub-Tier Design and Development Supplier List	E – Source Control Design Subcontractor List	See Section 6.3.6
Key Characteristics	F – Sensitive Source and Process List G – Sensitive Feature Drawing or List	See Section 6.3.2 , 6.3.15 and 6.3.16
Method of Manufacture	Retain on file at Supplier	See Section 6.3.14
Hazardous Material Definitions	H – Hazardous Material List	
Product Verification	V – Qualification Test Plans and Procedures	See Section 6.4
Verification Reports	J – Qualification Test Results	
Production Acceptance Tests	P – Acceptance Test Plans and Procedures	See Section 6.4.7
Production Acceptance Test Reports	M – Acceptance Test Results	
Repair Schemes	R – Repair Schemes	See Section 6.3.21
DFMEA and PFMEA with associated mitigations	T - DFMEA and PFMEA TD – DFMEA TP – PFMEA	See Section 6.3.2
Overall Statement of Compliance	S – Statement of Compliance	

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APPENDIX B – TECHNOLOGY READINESS LEVELS (TRL)

	Level	Description
SYSTEM VALIDATION	9	ACTUAL SYSTEM PROVEN THROUGH SUCCESSFUL MISSION OPERATION
	8	ACTUAL SYSTEM COMPLETED AND SERVICE QUALIFIED THROUGH TEST AND DEMONSTRATION
	7	SYSTEM PROTOTYPE DEMONSTRATION IN AN OPERATIONAL ENVIRONMENT
TECHNOLOGY VALIDATION	6	SYSTEM/SUBSYSTEM MODEL OR PROTOTYPE DEMONSTRATION IN A RELEVANT ENVIRONMENT
	5	COMPONENT AND/OR PARTIAL SYSTEM VALIDATION IN A RELEVANT ENVIRONMENT
APPLIED AND STRATEGIC RESEARCH	4	COMPONENT AND/OR PARTIAL SYSTEM VALIDATION IN A LABORATORY ENVIRONMENT
	3	ANALYTICAL AND EXPERIMENTAL CRITICAL FUNCTION AND/OR CHARACTERISTIC PROOF-OF-CONCEPT
	2	TECHNOLOGY CONCEPT AND/OR APPLICATION FORMULATED
	1	BASIC PRINCIPLES OBSERVED AND REPORTED

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APPENDIX C – MANUFACTURING CAPABILITY READINESS LEVELS (MCRL)

	Level	Description
PHASE 3 PRODUCTION IMPLEMENTATION	9	FULLY PRODUCTION CAPABLE PROCESS QUALIFIED ON FULL RANGE OF PARTS OVER EXTENDED PERIOD (ALL BUSINESS CASE METRICS ACHIEVED).
	8	FULLY PRODUCTION CAPABLE (FAIR STAGE 2) PROCESS QUALIFIED ON FULL RANGE OF PARTS OVER SIGNIFICANT RUN LENGTHS.
	7	CAPABILITY AND RATE CONFIRMED (FAIR STAGE 1 WITHOUT CONCESSIONS) VIA ECONOMIC RUN LENGTHS ON PRODUCTION PARTS.
PHASE 2 PRE-PRODUCTION	6	PROCESS OPTIMISED FOR CAPABILITY AND RATE USING PRODUCTION EQUIPMENT.
	5	BASIC CAPABILITY DEMONSTRATED USING PRODUCTION EQUIPMENT.
PHASE 1 TECHNOLOGY ASSESSMENT AND PROVING	4	PROCESS VALIDATED IN LABORATORY USING REPRESENTATIVE DEVELOPMENT EQUIPMENT.
	3	EXPERIMENTAL PROOF OF CONCEPT COMPLETED.
	2	APPLICABILITY AND VALIDITY OF CONCEPT DESCRIBED AND VETTED, OR DEMONSTRATED.
	1	PROCESS CONCEPT PROPOSED WITH SCIENTIFIC FOUNDATION.

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APPENDIX D– NOTIFICATION OF CHANGE (NOC) REQUIREMENTS

D1 The Supplier shall manage and control changes to the Design Technical Data Package in accordance with AS9116 using the applicable Rolls-Royce process. When the content of AS9116 or this Appendix conflicts with the Design and Development Quality Plan (ref Section 6.2.1.1), which has been approved by Rolls-Royce, the Design and Development Quality Plan takes precedence.

D2 Unless otherwise specified in the Design and Development Quality Plan, all changes listed in AS9116 section 4.2.1 and 4.2.2 shall be considered, and handled as "Prior Acceptance".

D3 The Supplier is allowed self-approval authority for changes listed in AS9116 as follows, provided that Rolls-Royce has formally granted such self-approval authority to the Supplier following the relevant Rolls-Royce process and the self-approval authority is documented in the Design and Development Quality Plan:

- Changes listed in AS9116 section 4.2.2 on an item by item basis
- Changes listed in AS9116 section 4.2.3 categorically
- Changes listed in AS9116 section 4.2.4 categorically

For changes where self-approval authority has been formally granted, the Supplier shall submit a NOC record using the Rolls-Royce change document (i.e. DAR) indicating Supplier self approval.

D4 The Supplier shall submit a NOC record to Rolls-Royce for approval using the applicable Rolls-Royce change document for changes that are not authorised as self-approval changes, as follows:

- For designs that were commissioned by Rolls-Royce UK, the Supplier shall use a Design Change Proposal (DCP) or Definition Alteration Request (DAR) form depending on the scope of the change.
- For designs that were commissioned by Rolls-Royce Deutschland, the Supplier shall use a Request for Alteration (RFA) or Definition Alteration Request (DAR) form, depending on the scope of the change.
- For designs that were commissioned by Rolls-Royce Corporation, the Supplier shall submit a Definition Alteration Request (DAR) form for all proposed changes.

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